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Application of auxiliary lighting in automatic hitch operation

Abstract

A vehicle hitching assistance system includes a controller executing an initial alignment process including searching for a trailer within a specified area of image data that is less than a total field of the image data in directions corresponding with both a longitudinal distance between the vehicle and the trailer and a lateral direction perpendicular to the longitudinal distance, presenting an indication to a driver of the vehicle to reposition the vehicle when the trailer is not identified within the specified area, and removing the indication when the trailer is identified within the specified area. The controller further executes an automated backing process upon identifying the trailer within the specified area, including identifying a coupler of the trailer and outputting a steering signal to the vehicle to cause the vehicle to steer to align a hitch ball of the vehicle with the coupler of the trailer during reversing.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present application is a continuation of U.S. patent application Ser. No. 17/950,237, now U.S. Pat. No. 11,964,692, filed on Sep. 22, 2022, entitled “APPLICATION OF AUXILIARY LIGHTING IN AUTOMATIC HITCH OPERATION,” which is a continuation of U.S. patent application Ser. No. 16/174,942, now U.S. Pat. No. 11,485,413, filed on Oct. 30, 2018, and entitled “APPLICATION OF AUXILIARY LIGHTING IN AUTOMATIC HITCH OPERATION,” the entire contents of which is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

(1) The present invention generally relates to a vehicle hitch assistance system. In particular, the system provides the user with various options for assisting in hitching a vehicle with a trailer and targets for initial alignment of the vehicle prior to assistance in hitching.

BACKGROUND OF THE INVENTION

(2) Hitching a trailer to a vehicle can be a difficult and time-consuming experience. In particular, aligning a vehicle hitch ball with the desired trailer hitch can, depending on the initial location of the trailer relative to the vehicle, require repeated forward and reverse driving coordinated with multiple steering maneuvers to appropriately position the vehicle. Further, through a significant portion of the driving needed for appropriate hitch ball alignment, the trailer hitch cannot be seen, and the hitch ball can, under ordinary circumstance, never actually be seen by the driver. This lack of sight lines requires inference of the positioning of the hitch ball and hitch based on experience with a particular vehicle and trailer, and can still require multiple instances of stopping and stepping out of the vehicle to confirm alignment or to note an appropriate correction for a subsequent set of maneuvers. Even further, the closeness of the hitch ball to the rear bumper of the vehicle means that any overshoot can cause a collision of the vehicle with the trailer. Accordingly, further advancements may be desired.

SUMMARY OF THE INVENTION

(3) According to one aspect of the disclosure, a vehicle hitching assistance system includes a controller acquiring image data from the vehicle and identifying a trailer within a specified area of the image data and then identifying a coupler of the trailer. The specified area is less than a total field of the image data. The controller then outputs a steering signal to the vehicle to cause the vehicle to steer to align a hitch ball of the vehicle with the coupler.

(4) Embodiments of the first aspect of the disclosure can include any one or a combination of the following features or aspects: the controller may acquire the image data from an imaging system included with the vehicle, the imaging system having at least one camera, the total field of the image data corresponding with a total field of view of the at least one camera; the controller may output the steering signal to a steering system included with the vehicle, and the controller may derive the steering signal based on at least a maximum steering angle of the steering system; the

specified area of the image data can be a target area disposed within a central portion of the image data; the specified area of the image data can be within a designated boundary comprising respective portions based on a resolution of the image data, a proportion of the trailer relative to the total field, and a known steering limit of the vehicle; the respective portions of the designated boundary may be based on a correlation of the total field of the image data with an area of an assumed ground plane on which the vehicle is positioned visible within the total field; the area of the assumed ground plane may include a maximum coupler detection distance corresponding with the resolution of the image data, a minimum trailer identification distance corresponding with the proportion of the trailer relative to the total field, and left and right maximum steerable paths extending from the vehicle in a reversing direction corresponding with the known steering limit of the vehicle; the controller may further output a video image displayable on a human-machine interface within the vehicle including the image data and a graphic overlay of the specified area on the image data in a proportionally correlated manner; the controller may output the graphic overlay in the video image upon activation of the system; the controller may receive an input from the human-machine interface corresponding with a user indication of a trailer within the image data and may output the graphic overlay in the video image only after receiving the user indication of the trailer within the image data and failing to identify a trailer within the specified area of the image data; the controller may further cause the vehicle to illuminate one or more exterior lights directed toward a rear of the vehicle prior to acquiring the image data from the vehicle; and the controller may identify the trailer within the specified area of the image data and then identify a coupler of the trailer and output a steering signal to the vehicle to cause the vehicle to steer to an align a hitch ball of the vehicle with the coupler as a part of a first hitch assist mode implemented when the controller determines that a sensing condition and a visibility condition are met, further implement a second hitch assistance mode when one of the sensing condition and the visibility condition are not met, receive a selection signal from the vehicle corresponding with a user selection of a mode before implementing either the first or second hitch assistance mode, and cause the vehicle to present an indication that the first hitch assist mode may not be selected when one of the sensing condition and the visibility condition are not met.

(5) According to another aspect of the disclosure, a vehicle includes a steering system, at least one exterior light mounted on and directed away from a rear of the vehicle, and a controller. The controller causes the at least one exterior light to illuminate, acquiring image data from the vehicle, identifies a trailer within the image data, and outputs a steering signal to the vehicle steering system to an align a hitch ball of the vehicle with a coupler of the trailer.

(6) According to another aspect of the disclosure, a method for assisting a vehicle in hitching with a trailer includes acquiring image data for a field of view away from a rear of the vehicle, identifying a trailer within a specified area less than the field of view of the image data and then identifying a coupler of the trailer, and outputting a steering signal to cause the vehicle to steer to an align a hitch ball of the vehicle with the coupler.

(7) According to another aspect of the disclosure, a vehicle hitching assistance system includes a controller acquiring image data from the vehicle and executing an initial alignment process. The initial alignment process includes searching for a trailer, positioned past a minimum distance from the vehicle, within a specified area of the image data, the specified area being less than a total field of the image data in directions corresponding with both a longitudinal distance between the vehicle and the trailer and a lateral direction perpendicular to the longitudinal distance and presenting an indication to a driver of the vehicle to reposition the vehicle when the trailer is not identified within the specified area and removing the indication when the trailer is identified within the specified area, the trailer remaining past the minimum distance from the vehicle. The controller further executes an automated backing process upon identifying the trailer within the specified area, including identifying a coupler of the trailer and outputting a steering signal to the vehicle to cause the vehicle to steer to align a hitch ball of the vehicle with the coupler of the trailer during

reversing of the vehicle toward the trailer.

(8) According to another aspect of the disclosure, a vehicle includes a steering system and a controller. The controller acquires image data from the vehicle and executes an initial alignment process, including searching for a trailer, positioned at a distance from the vehicle, within a specified area of the image data, the specified area being less than a total field of the image data in directions corresponding with both a longitudinal distance between the vehicle and the trailer and a lateral direction perpendicular to the longitudinal distance, the specified area further being located within an area to the rear of the vehicle illuminated by the at least one exterior light. An indication is presented to a driver of the vehicle to reposition the vehicle when the trailer is not identified within the specified area and removing the indication when the trailer is identified within the specified area, and an automated backing process is executed, only upon identifying the trailer within the specified area, and including outputting a steering signal to the vehicle steering system to align a hitch ball of the vehicle with a coupler of the trailer during reversing of the vehicle toward the trailer.

(9) According to another aspect of the disclosure, a method for assisting a vehicle in hitching with a trailer includes acquiring image data for a field of view away from a rear of the vehicle includes executing an initial alignment process, including searching for a trailer, past a minimum distance from the vehicle, within a specified area less than the field of view of the image data, the specified area being less than a total field of the image data in directions corresponding with both a longitudinal distance between the vehicle and the trailer and a lateral direction perpendicular to the longitudinal distance and presenting an indication to a driver of the vehicle to reposition the vehicle when the trailer is not identified within the specified area and removing the indication when the trailer is identified within the specified area, the trailer remaining past the minimum distance from the vehicle. The method further includes executing an automated backing process upon identifying the trailer within the specified area, including identifying a coupler of the trailer and outputting a steering signal to cause the vehicle to steer to align a hitch ball of the vehicle with the coupler during reversing of the vehicle toward the trailer.

(10) According to another aspect of the disclosure, a vehicle hitching assistance system includes a controller acquiring image data from the vehicle; searching for a trailer, positioned past a minimum distance from the vehicle, within a preselected area within the image data, the preselected area being less than a total field of the image data in directions corresponding with both a longitudinal distance between the vehicle and the trailer, by an amount corresponding with a maximum detection distance, a minimum detection distance corresponding with a proportion of the trailer relative to the total field, and a lateral direction perpendicular to the longitudinal distance by an amount corresponding with a known steering limit of the vehicle. The controller further determines that the trailer is not within the preselected area and executes an initial alignment process, including presenting an indication to a driver of the vehicle to reposition the vehicle such that the trailer is within the preselected area and removing the indication when the trailer is identified within the preselected area, the trailer remaining past the minimum distance from the vehicle. After completion of the initial alignment process, the controller executes an automated backing process, including identifying a coupler of the trailer and outputting a steering signal to the vehicle to cause the vehicle to steer to align a hitch ball of the vehicle with the coupler of the trailer during reversing of the vehicle toward the trailer.

(11) These and other aspects, objects, and features of the present invention will be understood and appreciated by those skilled in the art upon studying the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings:

- (2) FIG. 1 is a perspective view of a vehicle in an unhitched position relative to a trailer;
- (3) FIG. 2 is a diagram of a system according to an aspect of the disclosure for assisting in aligning the vehicle with a trailer in a position for hitching the trailer to the vehicle;
- (4) FIG. 3 is a depiction of an image received from a vehicle camera during an alignment sequence step with a target overlaid thereon;
- (5) FIG. 4 is an further depiction of an image received from a vehicle camera during a subsequent alignment sequence step with the target and additional information overlaid thereon;
- (6) FIG. 5 is an overhead schematic view of a vehicle during a step of the alignment sequence with the trailer;
- (7) FIG. 6 is a depiction of an image received from the vehicle camera during an alternative alignment sequence step with a various guidance paths overlaid thereon;
- (8) FIG. 7 is a flowchart depicting a sequence in presenting various alignment modes for selection by a user;
- (9) FIG. 8 is a depiction of a menu presentable to a user via a vehicle human-machine interface for selection of an alignment mode according to FIG. 7;
- (10) FIG. 9 is a schematic depiction of a valid zone for trailer identification based on alignment with a vehicle including a system according to FIG. 2;
- (11) FIG. 10 is a depiction of an image received from a vehicle camera during an alternative alignment sequence step with an alternative target overlaid thereon;
- (12) FIG. 11 is an further depiction of a subsequent image received from a vehicle camera during a subsequent alignment sequence step with the target and additional information overlaid thereon;
- (13) FIG. 12 is a depiction of an image received from a vehicle camera during an alternative alignment sequence step with an alternative target overlaid thereon;
- (14) FIG. 13 is a schematic depiction of an alternative valid zone for trailer identification based on alignment with a vehicle including a system according to FIG. 2;
- (15) FIG. 14 is a flowchart depicting a sequence in guiding a vehicle into alignment with a trailer for identification thereof by a system according to FIG. 2;
- (16) FIG. 15 is a depiction of an image received from a vehicle camera during the alignment sequence step of the sequence of FIG. 14 with an alternative target overlaid thereon;
- (17) FIG. 16 is a flowchart depicting an alternative sequence in guiding a vehicle into alignment with a trailer for identification thereof by a system according to FIG. 2;
- (18) FIGS. 17A and 17B are depictions of respective image received from a vehicle camera during alignment sequence steps of the sequence of FIG. 16;
- (19) FIG. 18 is an overhead schematic view of portions to the rear of a vehicle illuminated by various exterior lights included on the vehicle;
- (20) FIG. 19 is a flowchart depicting a sequence of selectively using exterior vehicle lights to improve the ability of a system according to FIG. 2 to detect a trailer;
- (21) FIG. 20 is an overhead schematic view of the vehicle during a subsequent step of the alignment sequence with the trailer;
- (22) FIG. 21 is a depiction of an image received from a vehicle camera during the alignment sequence step of FIG. 7;
- (23) FIG. 22 is an overhead schematic view of the vehicle during a subsequent step of the alignment sequence with the trailer;
- (24) FIG. 23 is an overhead schematic view of the vehicle during a subsequent step of the alignment sequence with the trailer and showing the position of a hitch ball of the vehicle at an end of a derived alignment path; and
- (25) FIG. 24 is a flowchart depicting steps in the alignment sequence.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(26) For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” “interior,” “exterior,” and derivatives thereof shall relate to the device as oriented in FIG. 1. However, it is to be understood that the device may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawing, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise. Additionally, unless otherwise specified, it is to be understood that discussion of a particular feature of component extending in or along a given direction or the like does not mean that the feature or component follows a straight line or axis in such a direction or that it only extends in such direction or on such a plane without other directional components or deviations, unless otherwise specified.

(27) Referring generally to FIGS. 1 and 2, reference numeral **10** designates a hitch assistance system (also referred to as a “hitch assist” system) for a vehicle **12**. In particular, hitch assistance system **10** includes a controller **26** a controller acquiring image data **55** from the vehicle **12** and identifying a trailer **16** within a specified area **45** of the image data **55** and then identifying a coupler **14** of the trailer **16**, the specified area **45** being less than a total field **53** of the image data **55**. The controller **26** further outputs a steering signal to the vehicle **12** to cause the vehicle **12** to steer to align a hitch ball **34** of the vehicle **12** with the coupler **34**.

(28) With respect to the general operation of the hitch assist system **10**, as illustrated in the system diagram of FIG. 2, system **10** includes various sensors and devices that obtain or otherwise provide vehicle status-related information. This information includes positioning information from a positioning system **22**, which may include a dead reckoning device **24** or, in addition or as an alternative, a global positioning system (GPS), to determine a coordinate location of the vehicle **12** based on the one or more locations of the devices within the positioning system **22**. In particular, the dead reckoning device **24** can establish and track the coordinate location of the vehicle **12** within a localized coordinate system **82** based at least on vehicle speed and steering angle δ . Other vehicle information received by hitch assist system **10** may include a speed of the vehicle **12** from a speed sensor **56** and a yaw rate of the vehicle **12** from a yaw rate sensor **58**. It is contemplated that in additional embodiments, a proximity sensor **54** or an array thereof, and other vehicle sensors and devices may provide sensor signals or other information, such as sequential images of a trailer **16**, including the detected coupler **14**, that the controller **26** of the hitch assist system **10** may process with various routines to determine the height H and position (e.g., based on the distance $D_{sub.h}$ and angle $\alpha_{sub.h}$) of coupler **14**.

(29) As further shown in FIG. 2, one embodiment of the hitch assist system **10** is in communication with the steering system **20** of vehicle **12**, which may be a power assist steering system **20** including an electric steering motor **74** to operate the steered wheels **76** (FIG. 1) of the vehicle **12** for moving the vehicle **12** in such a manner that the vehicle yaw changes with the vehicle velocity and the steering angle δ . In the illustrated embodiment, the power assist steering system **20** is an electric power-assisted steering (“EPAS”) system including electric steering motor **74** for turning the steered wheels **76** to a steering angle δ based on a steering command, whereby the steering angle δ may be sensed by a steering angle sensor **78** of the power assist steering system **20**. The steering command **69** may be provided by the hitch assist system **10** for autonomously steering during a trailer hitch alignment maneuver and may alternatively be provided manually via a rotational position (e.g., steering wheel angle) of a steering wheel of vehicle **12**. However, in the illustrated embodiment, the steering wheel of the vehicle **12** is mechanically coupled with the steered wheels **76** of the vehicle **12**, such that the steering wheel moves in concert with steered wheels **76**, preventing manual intervention with the steering wheel during autonomous steering. More specifically, a torque sensor **80** is provided on the power assist steering system **20** that senses

torque on the steering wheel that is not expected from autonomous control of the steering wheel and therefore indicative of manual intervention, whereby the hitch assist system **10** may alert the driver to discontinue manual intervention with the steering wheel and/or discontinue autonomous steering. In alternative embodiments, some vehicles have a power assist steering system **20** that allows a steering wheel to be partially decoupled from movement of the steered wheels **76** of such a vehicle.

(30) With continued reference to FIG. 2, the power assist steering system **20** provides the controller **26** of the hitch assist system **10** with information relating to a rotational position of steered wheels **76** of the vehicle **12**, including a steering angle δ . The controller **26** in the illustrated embodiment processes the current steering angle, in addition to other vehicle **12** conditions to guide the vehicle **12** along the desired path **32** (FIG. 3). It is conceivable that the hitch assist system **10**, in additional embodiments, may be an integrated component of the power assist steering system **20**. For example, the power assist steering system **20** may include a hitch assist algorithm for generating vehicle steering information and commands as a function of all or a portion of information received from the imaging system **18**, the power assist steering system **20**, a vehicle brake control system **70**, a powertrain control system **72**, and other vehicle sensors and devices, as well as a human-machine interface **40**, as discussed further below.

(31) As also illustrated in FIG. 2, the vehicle brake control system **70** may also communicate with the controller **26** to provide the hitch assist system **10** with braking information, such as vehicle wheel speed, and to receive braking commands from the controller **26**. For instance, vehicle speed information can be determined from individual wheel speeds as monitored by the brake control system **70**. Vehicle speed may also be determined from the powertrain control system **72**, the speed sensor **56**, and the positioning system **22**, among other conceivable means. In some embodiments, individual wheel speeds can also be used to determine a vehicle yaw rate $\{\dot{\gamma}\}$, which can be provided to the hitch assist system **10** in the alternative or in addition to the vehicle yaw rate sensor **58**. The hitch assist system **10** can, further, provide vehicle braking information to the brake control system **70** for allowing the hitch assist system **10** to control braking of the vehicle **12** during backing of the trailer **16**. For example, the hitch assist system **10**, in some embodiments, may regulate speed of the vehicle **12** during alignment of the vehicle **12** with the coupler **14** of trailer **16**, which can reduce the potential for a collision with trailer **16**, and can bring vehicle **12** to a complete stop at a determined endpoint **35** of path **32**. It is disclosed herein that the hitch assist system **10** can additionally or alternatively issue an alert signal corresponding to a notification of an actual, impending, and/or anticipated collision with a portion of trailer **16**. The powertrain control system **72**, as shown in the embodiment illustrated in FIG. 2, may also interact with the hitch assist system **10** for regulating speed and acceleration of the vehicle **12** during partial or autonomous alignment with trailer **16**. As mentioned above, regulation of the speed of the vehicle **12** may be advantageous to prevent collision with trailer **16**.

(32) Additionally, the hitch assist system **10** may communicate with human-machine interface (“HMI”) **40** for the vehicle **12**. The HMI **40** may include a vehicle display **44**, such as a center-stack mounted navigation or entertainment display (FIG. 1). HMI **40** further includes an input device, which can be implemented by configuring display **44** as a portion of a touchscreen **42** with circuitry **46** to receive an input corresponding with a location over display **44**. Other forms of input, including one or more joysticks, digital input pads, or the like can be used in place or in addition to touchscreen **42**. Further, the hitch assist system **10** may communicate via wireless communication with another embodiment of the HMI **40**, such as with one or more handheld or portable devices **96** (FIG. 1), including one or more smartphones. The portable device **96** may also include the display **44** for displaying one or more images and other information to a user. For instance, the portable device **96** may display one or more images of the trailer **16** on the display **44** and may be further able to receive remote user inputs via touchscreen circuitry **46**. In addition, the portable device **96** may provide feedback information, such as visual, audible, and tactile alerts.

(33) Still referring to the embodiment shown in FIG. 2, the controller **26** is configured with a microprocessor **60** to process logic and routines stored in memory **62** that receive information from the above-described sensors and vehicle systems, including the imaging system **18**, the power assist steering system **20**, the vehicle brake control system **70**, the powertrain control system **72**, and other vehicle sensors and devices. The controller **26** may generate vehicle steering information and commands as a function of all or a portion of the information received. Thereafter, the vehicle steering information and commands may be provided to the power assist steering system **20** for affecting steering of the vehicle **12** to achieve a commanded path **32** (FIG. 3) of travel for alignment with the coupler **14** of trailer **16**. The controller **26** may include the microprocessor **60** and/or other analog and/or digital circuitry for processing one or more routines. Also, the controller **26** may include the memory **62** for storing one or more routines, including an image processing **64** routine and/or hitch detection routine, a path derivation routine **66**, and an operating routine **68**. It should be appreciated that the controller **26** may be a stand-alone dedicated controller or may be a shared controller integrated with other control functions, such as integrated with a vehicle sensor system, the power assist steering system **20**, and other conceivable onboard or off-board vehicle control systems. It should further be appreciated that the image processing routine **64** may be carried out by a dedicated processor, for example, within a stand-alone imaging system for vehicle **12** that can output the results of its image processing to other components and systems of vehicle **12**, including microprocessor **60**. Further, any system, computer, processor, or the like that completes image processing functionality, such as that described herein, may be referred to herein as an “image processor” regardless of other functionality it may also implement (including simultaneously with executing image processing routine **64**).

(34) System **10** can also incorporate an imaging system **18** that includes one or more exterior cameras, which in the illustrated examples include rear camera **48**, center high-mount stop light (CMHSL) camera **50**, and side-view cameras **52a** and **52b**, although other arrangements including additional or alternative cameras are possible. In one example, imaging system **18** can include rear camera **48** alone or can be configured such that system **10** utilizes only rear camera **48** in a vehicle with multiple exterior cameras. In another example, the various cameras **48**, **50**, **52a**, **52b** included in imaging system **18** can be positioned to generally overlap in their respective fields of view, which may correspond with rear camera **48**, center high-mount stop light (CMHSL) camera **50**, and side-view cameras **52a** and **52b**, respectively. In this manner, image data **55** from two or more of the cameras can be combined in image processing routine **64**, or in another dedicated image processor within imaging system **18**, into a single image. In an extension of such an example, the image data **55** can be used to derive stereoscopic image data that can be used to reconstruct a three-dimensional scene of the area or areas within overlapped areas of the various fields of view **49**, **51**, **53a**, **53b**, including any objects (obstacles or coupler **14**, for example) therein. In an embodiment, the use of two images including the same object can be used to determine a location of the object relative to the two image sources, given a known spatial relationship between the image sources. In this respect, the image processing routine **64** can use known programming and/or functionality to identify an object within image data **55** from the various cameras **48**, **50**, **52a**, and **52b** within imaging system **18**. In either example, the image processing routine **64** can include information related to the positioning of any cameras **48**, **50**, **52a**, and **52b** present on vehicle **12** or utilized by system **10**, including relative to the center **36** (FIG. 1) of vehicle **12**, for example such that the positions of cameras **48**, **50**, **52a**, and **52b** relative to center **36** and/or to each other can be used for object positioning calculations and to result in object position data relative to the center **36** of vehicle **12**, for example, or other features of vehicle **12**, such as hitch ball **34** (FIG. 1), with known positions relative to center **36**.

(35) The image processing routine **64** can be specifically programmed or otherwise configured to locate coupler **14** within image data **55**. In the example of FIGS. 3 and 4, the image processing routine **64** can first attempt to identify any trailers **16** within the image data **55**, which can be done

based on stored or otherwise known visual characteristics of trailer **16**, of an number of different types, sizes or configurations of trailers compatible with system **10**, or trailers in general. When a trailer **16** is identified, system **10** may cause an indication of such identification to be presented to the user via the vehicle HMI, including the box **17** shown in FIG. **4**, which can be superimposed on the image presented on HMI **40** by or based on an output from controller **26**. In connection with such an identification and indication **17** of trailer **16**, controller **26** can seek confirmation from the user that the identification of the trailer **16** is accurate and is the correct trailer for which to complete an automated hitching operation, as described further below. In the illustrated example, a graphical button **90** can be presented on HMI **40** adjacent the trailer indication box **17** and the user can be requested to confirm the trailer identification before controller **26** proceeds with the automated hitching operation.

(36) After the trailer **16** is identified, controller **26** may then identify the coupler **14** of that trailer **16** within the image data **55** based, similarly, on stored or otherwise known visual characteristics of coupler **14** or couplers in general. In another embodiment, a marker in the form of a sticker or the like may be affixed with trailer **16** in a specified position relative to coupler **14** in a manner similar to that which is described in commonly-assigned U.S. Pat. No. 9,102,271, the entire disclosure of which is incorporated by reference herein. In such an embodiment, image processing routine **64** may be programmed with identifying characteristics of the marker for location in image data **55**, as well as the positioning of coupler **14** relative to such a marker so that the location **28** of coupler **14** can be determined based on the marker location. Additionally or alternatively, controller **26** may seek confirmation of the determined coupler **14**, via a prompt on touchscreen **42** similar to the box **17** used to prompt for confirmation the trailer **16**. If the coupler **14** determination is not confirmed, further image processing may be provided, or user-adjustment of the position **28** of coupler **14** may be facilitated, either using touchscreen **42** or another input to allow the user to move the depicted position **28** of coupler **14** on touchscreen **42**, which controller **26** uses to adjust the determination of position **28** of coupler **14** with respect to vehicle **12** based on the above-described use of image data **55**. In various examples, controller **26** may initially rely on the identification of trailer **16** for the initial stages of an automated hitching operation, with the path **32** being derived to move the hitch ball **34** toward a centrally-aligned position with respect to trailer **16** with the path **32** being refined once the coupler **14** is identified. Such an operational scheme can be implemented when it is determined that trailer **16** is at a far enough distance from vehicle **12** to begin backing without knowing the precise endpoint **35** of path **32** and can be useful when trailer **16** is at a distance where the resolution of the image data **55** makes it possible to accurately identify trailer **16**, but at which the coupler **14** cannot be precisely identified. In this manner, initial rearward movement of vehicle **12** can allow for calibration of various system **12** inputs or measurements that can improve the accuracy of distance measurements, for example, that can help make coupler **14** identification more accurate. Similarly, movement of vehicle **12** resulting in a change to the particular image within the data **55** that can improve the resolution or move the coupler **14** relative to the remaining portions of trailer **16** such that it can be more easily identified.

(37) In this manner, the initial determination of the position **28** of trailer **16** to an accepted level of accuracy is needed for execution of the path derivation routine **66** and subsequent automated backing of vehicle **12** along the path **32**. Various characteristics or limitations of system **10** may impact the ability of system **10** to identify the trailer **16** (as well as the coupler **14**, whenever such identification is carried out) in the data **55** received from imaging system **18** under certain conditions or in certain settings. Still further, various vehicle **12** or other system **10** characteristics may impact the ability of system **10** to navigate to reach a trailer **16** that is, nevertheless, present within the image data **55**. Depending on the particular configuration of system **10**, such characteristics can be partially driven by the imaging system **18** used by system **10**. The imaging system **18** may be limited in its ability to identify a trailer **16** and/or coupler **14** within the entire field of the image data **55**. In an example, it may be assumed, at least for simplicity of illustration,

that system **10** only uses rear camera **48** for trailer **16** and coupler **14** detection, with rear camera **48** having a field of view **49** that is included in its entirety in the “total field” of the image data **55** (notably, if additional cameras **50, 52a, 52b** are used, the total field of the image data **55** would include the entire assembled image from all such utilized cameras). The imaging system **18** limitations may limit system **10** functionality to only a limited distance between trailer coupler **14** and the vehicle **12**, as different factors may limit the ability of controller **26** in identifying a trailer **16** or its coupler **14** when the trailer **16** and vehicle **12** are too close together or too far apart. For example, the resolution of the various cameras **48, 50, 52a, 52b** in imaging system **18** may impact the ability to identify any trailers **16** or couplers **14** beyond a maximum distance **R1** from vehicle **12** with the particular value of **R1** being influenced by ambient conditions, including available light and/or weather conditions (e.g., rain or snow).

(38) Additionally, a minimum distance **R2** for trailer **16** or coupler **14** detection may be realized because certain implementations of system **10** may rely on dynamic readings (such as of the ground surface behind vehicle **12** or other features visible around coupler **14**) to calibrate system **10** and or to track vehicle **12** speed in reversing and to track the position of coupler **14** during system **10** operation. In particular, in the above example where only rear camera **48** is used by system **10**, it may be necessary to detect motion within the field of view **49** to identify distance to the coupler **14** and to provide accurate tracking and boundary resolution (an aspect of image processing routine **64**). Further, the operating routine **68** may include a longitudinal control algorithm that relies on precise control of the vehicle **12**, and a minimum amount of travel distance corresponding with **R2** in an example, is required to calibrate certain braking and powertrain variables to achieve such vehicle control. Still further, if a trailer **16** is too close to vehicle **12**, various features of the trailer **16** may appear as trailers themselves to the image processing routine **64**, meaning that to assist system **10**, the trailer **16** should be beyond the minimum distance **R2** such that a proportionality of features, including of trailer **16** itself as well as of trailer **16** relative to the total field of image data **55**, is optimized for image processing routine **64** functionality.

(39) Additionally, other limitations of system **10** functionality may add constraints to the acceptable zone of operation. In this respect, system **10** may not be capable of maneuvering vehicle **12** towards all locations in an initial view of the rear camera **48** (i.e., during trailer **16** or coupler **14** identification). In particular, system **10** is restricted in its ability to reach a potential target position due, but not limited, to a lateral span that is a function of a distance range and the steering angle δ limitations of vehicle **12**. In one aspect, the maximum steering angle $\delta_{sub.max}$ of the vehicle **12** determines the lateral range, as a function of distance $D_{sub.c}$ to coupler **14**, as discussed further below. In general, an implementation of system **10** may restrict maneuvering of vehicle **12** to a single reversing motion that, while potentially including steering in both the left and right directions, does not incorporate forward driving of vehicle **12** between successive instances of reverse driving, for example. In this manner, the maximum lateral distance that can be traversed by vehicle **12** in an automated hitching operation is limited by the maximum steering angle $\delta_{sub.max}$. As the vehicle **12** travels laterally by turning the steered wheels **76** and reversing, the lateral limits of system operability **10** are determined as, essentially, a theoretical hitch ball **34** path extending rearward of the vehicle corresponding with steering of vehicle **12** at the maximum steering angle under reversing of vehicle to either side. In this manner, the lateral limits of system **10** may extend outwardly from vehicle **12**, with increasing distance away from vehicle **12**. In a further aspect, the steering angle δ may be limited to an angle δ_a that is lower than maximum steering angle $\delta_{sub.max}$ based on predetermined constraints for allowable swing of the front end of vehicle **12**. In this manner, the lateral limits of system **10** functionality may be further limited.

(40) Because of these limitations, the present system **10** may be configured to only function with trailers **16** and associated couplers **14** positioned inside a “valid” region of space relative to the vehicle **12**. The region is determined by the factors listed above, and, potentially, any additional factors that affect the system **10** capability. To ensure such positioning of vehicle **12** relative to

trailer **16**, system **10** can be generally configured to direct the user to position vehicle **12** relative to trailer **16** such that trailer **16** is within such a valid area of the field of view of the utilized cameras, such as field of view **49** of rear camera **48**, and the corresponding image data **55**. As shown in FIGS. **3** and **4**, this direction may be given by way of presenting a target **45** as a graphical overlay on a real-time video image **53** of the image data **55** from one or more of the cameras **48**, **50**, **52a**, **52b** in imaging system **18** presented on screen **44**. The target **45** may be derived and/or presented on screen **44** according to various characteristics of system **10** described above and may balance these characteristics and system requirements to provide widely useable functionality of system **10**, overall. In this manner, the target **45** be positioned within the image **53** in a location that is determined to correspond with an actual location relative to vehicle **12** relative to the ground plane **30** on which vehicle **12** is positioned (and on which trailer **16** may be assumed to be present, regardless of actual ground characteristics) that is within the valid zone for trailer **16** and coupler **14** detection and vehicle **12** navigation for alignment therewith. In the example show, the target **45** may not directly correspond with the complete area within which such detection and navigation is possible, but may rather be a general area that is reliably within the acceptable zone, and requires placement of the coupler **14** and/or trailer **16** within a certain distance from vehicle **12**, including within a maximum and minimum distance from vehicle **12**, as well as within a predetermined maximum lateral offset from vehicle **12**. As shown, this may result in the target **45** being positioned generally centrally within the image **53** in both the horizontal and vertical directions and may represent, for example about 5-15% of the total area of image **53**. In some instances, target **45** may not be exactly centered within image **53**, at least in the vertical direction, with target **45** potentially being centered between about 30% and 50% of the vertical distance of image **54**. In various examples, the positioning of the particular camera(s), such as camera **48**, on vehicle **12**, as well as the characteristics (focal length, etc.) of the camera(s).

(41) When initiated, system **10** can automatically attempt to identify a trailer **16** within the area of target **45** while prompting the driver to position vehicle **12** such that the coupler **14** and/or trailer **16** is within the area of target **45**. When a trailer **16**, including its coupler **14**, are detected (which would generally coincide with positioning thereof within the area of target **45**, system **10** can indicate such an identification, as discussed above, by highlighting the trailer with box **17** (FIG. **4**), while instructing the driver to confirm (by pressing button **90**, for example) to confirm that the desired trailer **16** has been identified or to select the target trailer **16** from one or more identified trailers **16**. At which point, vehicle **12**, according to various potential interactive schemes, can acquire control of vehicle **12** from the user and can control vehicle **12** in aligning hitch ball **34** with coupler **14** for hitching of vehicle **12** with the trailer **16**.

(42) As shown in FIG. **5**, the image processing routine **64** and operating routine **68** may be used in conjunction with each other to determine the path **32** along which hitch assist system **10** can guide vehicle **12** to align hitch ball **34** and coupler **14** of trailer **16**. Upon initiation of hitch assist system **10**, such as by user input on touchscreen **42**, for example, image processing routine **64** can identify coupler **14** within the image data **55** and at least attempt to estimate the position **28** of coupler **14** relative to hitch ball **34** using the image data **55** in accordance with one of the examples discussed above to determine a distance $D_{sub.c}$ to coupler **14** and an angle $\alpha_{sub.c}$ of offset between coupler **14** and the longitudinal axis of vehicle **12**. Image processing routine **64** can also be configured to identify the trailer **16** overall and can use the image data of trailer **16**, alone or in combination with the image data of coupler **14**, to determine the orientation or heading **33** of trailer **16**. In this manner the path **32** can further be derived to align vehicle **12** with respect to trailer **16** with the longitudinal axis **13** of vehicle **12** within a predetermined angular range of the heading **33** of trailer **16**. Notably, such alignment may not require that the longitudinal axis **13** of vehicle **12** is parallel or collinear with the heading **33** of trailer **16**, but may simply be within a range that generally allows connection of hitch ball **34** with coupler **14** without collision between vehicle **12** and trailer **16** and may, further allow immediate controlled backing of trailer **16** using vehicle **12**. In this

manner, the angular range may be such that the alignment of vehicle **12** with trailer **16** at the end of the operating routine **68** is such that the angle between longitudinal axis **13** and heading **33** is less than the jackknife angle between the vehicle **12** and trailer **16** when coupled or a reasonable estimate thereof. In one example, the angular range may be such that longitudinal axis **13** is within about 30° from collinear with heading **33** in either direction. In various examples, such as when the length L of trailer **16** is known, the angular range may be greater, when permitted, or may be less, depending on the desired tolerance of system **10**.

(43) When collected, the position information can then be used in light of the position **28** of coupler **14** within the field of view of the image data **55** to determine or estimate the height H.sub.c of coupler **14**. Once the positioning D.sub.c, α .sub.c of coupler **14** has been determined and, optionally, confirmed by the user, controller **26** can take control of at least the vehicle steering system **20** to control the movement of vehicle **12** along the desired path **32** to align the vehicle hitch ball **34** with coupler **14**, as discussed further below.

(44) Continuing with reference to FIG. 5 with additional reference to FIG. 2, controller **26**, having estimated the positioning D.sub.c, α .sub.c of coupler **14**, as discussed above, can, in one example, execute path derivation routine **66** to determine vehicle path **32** to align the vehicle hitch ball **34** with coupler **14**. In particular, controller **26** can have stored in memory **62** various characteristics of vehicle **12**, including the wheelbase W, the distance from the rear axle to the hitch ball **34**, which is referred to herein as the drawbar length L, as well as the maximum angle to which the steered wheels **76** can be turned δ .sub.max. As shown, the wheelbase W and the current steering angle δ can be used to determine a corresponding turning radius ρ for vehicle **12** according to the equation:

$$(45) \quad \rho = \frac{1}{W \tan \delta}, \quad (1)$$

in which the wheelbase W is fixed and the steering angle δ can be controlled by controller **26** by communication with steering system **20**, as discussed above. In this manner, when the maximum steering angle δ .sub.max is known, the smallest possible value for the turning radius ρ .sub.min is determined as:

$$(46) \quad \rho_{\min} = \frac{1}{W \tan \delta_{\max}}. \quad (2)$$

(47) Path derivation routine **66** can be programmed to derive vehicle path **32** to align a known location of the vehicle hitch ball **34** with the estimated position **28** of coupler **14** that takes into account the determined minimum turning radius ρ .sub.min to allow path **32** to use the minimum amount of space and maneuvers. In this manner, path derivation routine **66** can use the position of vehicle **12**, which can be based on the center **36** of vehicle **12**, a location along the rear axle, the location of the dead reckoning device **24**, or another known location on the coordinate system **82**, to determine both a lateral distance to the coupler **14** and a forward or rearward distance to coupler **14** and derive a path **32** that achieves the needed lateral and forward-backward movement of vehicle **12** within the limitations of steering system **20**. The derivation of path **32** further takes into account the positioning of hitch ball **34**, based on length L, relative to the tracked location of vehicle **12** (which may correspond with the center **36** of mass of vehicle **12**, the location of a GPS receiver, or another specified, known area) to determine the needed positioning of vehicle **12** to align hitch ball **34** with coupler **14**. It is noted that hitch assist system **10** can compensate for horizontal movement Δx of coupler **14** in a driving direction away from axle **84** by determining the movement of coupler **14** in the vertical direction Δy that will be needed to receive hitch ball **34** within coupler **14**. Such functionality is discussed further in co-pending, commonly-assigned U.S. patent application Ser. Nos. 14/736,391 and 16/038,462, the entire disclosures of which are hereby incorporated by reference herein.

(48) As discussed above, once the desired path **32**, including endpoint **35**, has been determined using either of the offset determination schemes discussed above, controller **26** is then allowed to at least control the steering system **20** of vehicle **12** with the powertrain control system **72** and the brake control system **70** (whether controlled by the driver or by controller **26**, as discussed below)

controlling the velocity (forward or rearward) of vehicle **12**. In this manner, controller **26** can receive data regarding the position of vehicle **12** during movement thereof from positioning system **22** while controlling steering system **20**, as needed to maintain vehicle **12** along path **32**. In particular, the path **32**, having been determined based on the vehicle **12** and the geometry of steering system **20**, can adjust the steering angle δ , as dictated by path **32**, depending on the position of vehicle **12** there along. It is additionally noted that in an embodiment, the path **32** may comprise a progression of steering angle δ adjustment that is dependent on the tracked vehicle position.

(49) As illustrated in FIG. 5, vehicle path **32** can be determined to achieve the needed lateral and rearward movement within the smallest area possible and/or with the lowest number of maneuvers. In the illustrated example of FIG. 5, path **32** can include two portions defined by steering of wheels **76** in different directions to collectively traverse the needed lateral movement of vehicle **12**, while providing final straight, rearward backing segment to bring hitch ball **34** into the above-described offset alignment with coupler **14**. It is noted that variations in the depicted path **32** may be used. It is further noted that the estimates for the positioning D_e , a_c of coupler **14** may become more accurate as vehicle **12** traverses path **32**, including to position vehicle **12** in front of trailer **16** and as vehicle **12** approaches coupler **14**. Accordingly, such estimates can be continuously derived and used to update path derivation routine **66**, if necessary, in the determination of the adjusted endpoint **35** for path **32**, as discussed above. In a similar manner, the path **32**, as derived using the position and orientation data acquired from smartphone **96**, can be fine-tuned once the image processing routine **64** can identify coupler **14** in the image data **55**, with continued updates for path **32** being similarly derived as the image data **55** becomes increasingly clear during the approach toward trailer **16**. It is further noted that, until such a determination can be made, the dead reckoning device **24** can be used to track the location of vehicle **12** in its movement along path **32** toward the initially-derived endpoint **35**.

(50) As discussed above, system **10** requires the availability of a number of measurements obtained using imaging system **18** and, optionally, various sensors **54** and devices **22**, as well as reliable control of the steering, **20**, powertrain **72** and braking **70** systems to implement the image processing **64**, path derivation **66**, and operating routines **68** for control the backing of vehicle **12** according to the process described above. Accordingly, the inability of system **10** to obtain any such measurements or to reliably control any of the involved vehicle systems can impact the ability of system **10** to reliably carry out the above hitching process. Accordingly, system **10** may also be configured to provide multiple levels of hitching assistance depending on both user preference, as well as measurement availability and control reliability. For example, a user may not feel comfortable relinquishing control (completely or at all) of vehicle **12**, but may still prefer some level of guidance in aligning hitch ball **26** with coupler **14**. In other examples, visibility by way of available light or weather may impact the ability of system **10** locate or track coupler **14**, even when trailer **16** is within the above-described acceptable zone, or system **10** may determine that for various reasons, the steering **20** and braking **70** systems cannot be reliably controlled. Generally, control of the steering **20** system may be impacted by vehicle **12** being positioned on a transverse slope, which may cause wheel slip to cause vehicle **12** to travel on an unexpected path, and moisture, for example, may cause the brake system **70** functionality. Still further, positioning of vehicle **12** on soft ground or on an upward or downward slope may make powertrain **72** or braking **70** system control operate out of an optimal range. These or other conditions may diminish to the point where reliable control by system **10** is not available.

(51) To address any of the above, or other similar potential, scenarios, system **10** may include additional functionality according to FIGS. 6-8. In one example, system **10** may be configured to additionally operate in a “basic” mode, where various vehicle paths are displayed on screen **42** as an overlay on the image data **55**. As shown, the paths may include a vehicle path **92** that extends rearward of approximations of the rear corners of vehicle **12** and adjusts dynamically based on the

steering angle δ detected by steering angle sensor 78. The paths may further include a hitch ball path 94 that extends generally from the center of the rear of vehicle 12 so as to align with the hitch ball 34 of vehicle 12 and to show a projected path of the hitch ball 34, given the detected steering angle δ . In addition, system 10 may also overlay a braking indication image 96 that can signal to the user when system 10 determines, such as by use of proximity sensors 54, that the coupler 14 is in a position relative to vehicle 12 to be at least longitudinally aligned with hitch ball 34. These paths 92, 94 are based on calculations regarding the steering angle δ and, therefore, are not susceptible to the visibility requirements discussed above for the various operating routines 64, 66, 68. Similarly, the proximity sensors 54 may be ultrasonic or the like and, accordingly, do not require any ambient light to function and may be less susceptible diminished functionality due to weather conditions. Further, the presentation of such paths 92, 94, and the braking image 98 on image 53 allows for continuing assessment and adjustment by the driver should wheel slip or diminished braking functionality result in hitch ball 34 not following path 94, exactly. Accordingly, such a “basic” mode may be available under all conditions, subject to an assessment by the user that the coupler 14 is visible to the user on display 42.

(52) In an additional level of functionality, when coupler 14 can be identified by system 10, but steering system 20, braking system 70 and powertrain system 72 cannot be controlled with acceptable reliability, system may offer an “ideal” path 100 based on the user of the image processing 64 and path derivation 66 routines that can represent a path 32 determined by system 10 to align hitch ball 34 with coupler 14. When present, the user can adjust the steering input for vehicle 12 such that the hitch ball path 94 aligns with the ideal path 98 during reversing, while also controlling steering and braking until system 10 presents the braking indication 96 or the driver determines that appropriate longitudinal positioning has been achieved. Still further, system 10 may provide automatic steering of vehicle 12 by control of steering system 20 to maintain vehicle 12 along the determined path 32 while the user controls the operation of powertrain 72 and braking 70 system to control the speed of vehicle 12. This functionality can be used, for example, where visibility or ground conditions, for example allow for coupler 14 detection, but not ground tracking, or where powertrain 72 and/or brake system 70 cannot be controlled with the required accuracy, as discussed above. Finally, when the above-described conditions are met, the operability described above, including full control of vehicle 12 can be achieved.

(53) System 10, when providing the various levels of functionality discussed above, can additionally operate according to the scheme 200 depicted in the flow chart of FIG. 7, wherein, upon activation 202 of system 10, the various hitch assist modes are presented 204 on the HMI 40 as selectable options respectively corresponding with the “basic” 102, “guided path” 104, automated steering 106, and full-control 108 modes discussed above. In one example, when presenting 202 such options, system 10 can first assess 204 availability of the various control modes, according to an assessment of the various characteristics discussed above, and can only present available options, or can display all options, with any unavailable options being indicated as un-selectable (such as by being dimmed, crossed out, greyed out, or the like). The system 10 can then receive 206 a mode selection from the user, which may be stored as a default mode or may be single-instance selection, before activating 208 the selected functionality according to the selected mode, which may remain the mode for subsequent operations until changed 210 by the user.

(54) Turning to FIGS. 9-17B, various additional examples of executing the above-described full control hitch assist functionality by guiding the user to an initial vehicle 12 alignment with the subject trailer 16 in an initial acceptable zone are described. In one example, shown particularly in FIGS. 9-13, the target 45 may be displayed in a shape that more directly corresponds with the actual acceptable zone for trailer 16 and coupler 14 positioning relative to vehicle 12, which may provide increased flexibility and greater understanding of the system 10 requirements for the user compared to the generalized target of FIGS. 3 and 4. As discussed above, the visibility requirements of the operating routines 64, 66, 68 may dictate that trailer 16 (or at least coupler 14)

be positioned between a longitudinal range between limits R1 and R2 that correspond with distance from the vehicle 12, as shown in FIG. 9. As also shown in FIG. 9, the lateral range for the acceptable zone extends between the lateral limits L1 and L2, which as discussed above extend outwardly from the hitch ball 34 along “maximum steerable” paths corresponding with a maximum or maximum allowed steering angle $\delta_{sub.max}$ or $\delta_{sub.a}$ in both the left and right directions. In this manner, the acceptable zone 110 is an area along the ground plane 30 that is within both the longitudinal range and the lateral range and is, therefore, an area bounded by respective portions of the longitudinal limits R1,R2 and the lateral limits L1, L2.

(55) As shown in FIG. 10, in an example, the target 45 can be presented on screen 42 as an overlay on the video image 53 displayed using the image data 55 that approximates the acceptable zone 110 on the image 53. In this manner, the respective portions of the ranges R1, R2, L1, L2 bounding the acceptable zone 110 can be graphically represented on screen 42 in a manner that correlates the acceptable zone 110 on the actual ground plane 30 with the view on the screen 42 based on the properties of camera 48, for example, and the position thereof to arrive at a perspective projection of the acceptable zone that at least reasonably appears as an area of the ground visible on the screen 42. In this manner, the user may position vehicle 12 such that the subject trailer is within the target 45 corresponding with the acceptable zone 110. In the example illustrated in FIG. 11, once such positioning is achieved and system 10 detects at least trailer 16 within the acceptable zone 110 or aligned with target 45, the indication 17 of trailer 16 identification can be overlaid on trailer 16 within image 53 and button 90 can be displayed for user confirmation of the intended trailer 16. In an alternative confirmation scheme, depicted in FIG. 12, system 10 can seek confirmation based on user action other than a touch on HMI 40. As shown, system 10 can present a prompt 108 on display 42 for the user to stop the vehicle 12 to initiate the hitching operation.

(56) As shown in FIGS. 13-17B, positioning of vehicle 12 with trailer 16 positioned in the acceptable zone 110 can be “coached” by system 10 depicting the target 45 as, essentially, a void area within a depicted invalid zone 112 overlaid on image 53, as shown in FIG. 13. It is noted that the invalid zone 112 can simply be an inverse of the target 45 depicting the acceptable or “valid” zone 110, shown in FIGS. 10 and 12. For illustration purposes, the invalid zone 112 is depicted differently to illustrate the effect of varying system 10 constraints that can result in a larger valid zone 110, as shown in FIG. 14. In one aspect, system 10 can utilize the invalid zone 112 in a similar manner to the valid zone 110 target 45 discussed above with respect to FIGS. 9-12. In another example, system 10 can operate according to the scheme 300 depicted in FIG. 14, in which upon activation 302, system 10 causes the invalid zone 112 image to be presented 304 on display 42 as an overlay on the image 53 of the data 55 from camera 48, for example. The system 10 then relies on the user to assess 306 whether the trailer 16 is in a valid position (i.e. outside of the invalid zone 112) and to correct 308 the vehicle position, if needed. When the positioning is correct, the user can indicate 310 the trailer position by a touch input on screen 42, which the system 10 can use to narrow the field for image processing routine 64, which can then be used to identify 312 trailer 16 (and coupler 14 in an example) before ending the selection routine 314.

(57) In another example, system 10 can operate by the scheme 400 of FIG. 16, wherein, upon activation 402, system 10 waits for a user indication 402 of the trailer 16 position via a touch input on screen 42 (as shown in FIG. 17A). If possible, the system 10 can identify 406 the trailer 16 using image processing routine 64 in connection with the touch input. If the trailer 16 is identified, system 10 can analyze 408 the trailer 16 position and can determine 410 whether trailer 16 is in the valid zone 110. If the trailer 16 is not in the valid zone 110 (or cannot be identified, indicating, potentially, that the trailer is outside of the longitudinal range), system 10 can then present 412 the invalid zone and instruct 414 the driver to reposition 416 the vehicle 12 accordingly, while maintaining 418 the tracking of the trailer 16 (if detected) until trailer 16 is in the valid zone 110. When trailer 16 is identified within the target zone 110, system 10 can end the identification scheme and can proceed to the available guidance operation, as discussed above.

(58) As shown in FIG. 18, system 10 may utilize the various exterior lights on and/or directed away from the rear of vehicle 12 to illuminate the area behind vehicle 12 to improve the ability of system 10 to locate and track trailer 16 and coupler 14 during the operation of at least the image processing routine 64 and operating routine 68. As illustrated, such lights may include the right 114a and left 114b brake lights, the center high-mount stop light (“CMHSL”) 116, and the tailgate light 118, and may vary in the particular arrangement and lights included, depending on the vehicle 12 configuration. In this manner, more or fewer lights may be present with vehicle 12, for example, potentially further including backup lights, taillights, etc. In the illustrated example, system 10, upon activation can illuminate the depicted right 114a and left 114b brake lights, CMHSL 116, and the tailgate light 118, regardless of detected conditions or may analyze the image data 55 to determine the lighting conditions or to make an initial assessment regarding potential impact of weather conditions on the ability to identify one or more trailers 16 within the field of view 49 of camera 48, for example. Such illumination can be initiated by system 10 by way of communication with the vehicle lighting control system or functionality. As shown in FIG. 19, in one implementation, system 10 may operate 500 by a process including initially detecting the ambient lighting level, either using the camera 48 (for example) or using other vehicle sensors (steps 502a or 502b). System 10 can then determine 504 whether the available lighting is sufficient and proceed 506 with the automated hitching operation (as discussed above) or may activate 508, 510 the available exterior lighting before performing 512 the hitching operation with the additional lighting and deactivating 514a, 514b the exterior lighting (or allowing it to return to an initial state) when the operation is complete.

(59) The additional illumination provided by the illustrated lights 114a, 114b, 116, 118 may facilitate the identification by system 10 of any trailers 16 within the field of view 49 of camera 48, for example, by improving the contrast and/or resolution of the image data 55 available to system 10, particularly in the area where the illumination of such lights overlaps. As shown in FIG. 18, in some vehicle 12 and lighting configurations, the area where the illumination areas 120a, 120b, 122, 124 (of lights 114a, 114b, 116, 118, respectively) overlap may wholly or partially coincide with all or part of the acceptable zone 110, thereby providing the ability of system 10 to reliably utilize the target 45 discussed above for vehicle 12 and trailer 16 alignment in an increase range of ambient conditions, including those with low ambient light, and the like. In this manner, the acceptable zone 110 and resulting target configuration 45 may dynamically adjust in size or configuration (particularly with respect to the longitudinal range limits R1, R2) depending on the available light and/or the use of the exterior lighting during image processing routine 64. In another example, target 45 can be sized and configured based on limits R1, R2 during low-light conditions augmented by exterior light illumination to more reliably ensure system 10 reliability without requiring prior condition detection.

(60) Turning now to FIGS. 20-23, once the trailer 16 and coupler 14 have been identified, and system 10 determines the path 32 to align hitch ball 34 with the coupler 14, the operating routine 68 may continue to guide vehicle 12 until hitch ball 34 is in the desired position 38a relative to coupler 14 for coupler 14 to engage with hitch ball 34 when coupler 14 is lowered into horizontal alignment therewith. In the example discussed above, image processing routine 64 continuously monitors the positioning D.sub.c, α .sub.c of coupler 14, constantly or once available, during execution of operating routine 68, including as coupler 14 comes into clearer view of rear camera 48, as shown in FIG. 21, with continued movement of vehicle 12 along path 32, as shown in FIG. 20. As discussed above, the position of vehicle 12 can also be monitored by dead reckoning device 24 with the position 28 of coupler 14 being continuously updated and fed into path derivation routine 66 in case path 32 and or endpoint 35 can be refined or should be updated (due to, for example, improved height H.sub.c, distance D.sub.c, or offset angle α .sub.c information due to closer resolution or additional image data 55), including as vehicle moves closer to trailer 16, as shown in FIG. 22. Still further, the coupler 14 can be assumed to be static such that the position of

vehicle **12** can be tracked by continuing to track the coupler **14** to remove the need for use of the dead reckoning device **24**. In a similar manner, a modified variation of operating routine **68** can progress through a predetermined sequence of maneuvers involving steering of vehicle **12** at or below a maximum steering angle $\delta_{\text{sub.max}}$, while tracking the position $D_{\text{sub.c}}$, $\alpha_{\text{sub.c}}$ of coupler **14** to converge the known relative position of hitch ball **34** to the desired position **38d** thereof relative to the tracked position **28** of coupler **14**, as discussed above and shown in FIG. **23**.

(61) Turning now to FIG. **24**, a flowchart showing steps in one operating scheme **600** for using hitch assist system **10** to align a vehicle hitch ball **34** with a trailer coupler **14** is shown. In particular, in step **602**, the hitch assist system **10** is initiated. Once the hitch assist system **10** is initiated **602**, controller **26** can use imaging system **18** to scan the viewable scene using any or all available cameras **48**, **50**, **52a**, **52b** (step **604**). The scene scan (step **612**) can be used to then identify **606** the trailer **16** and coupler **14**, which may be confirmed by the user, which may proceed using target **45** on vehicle HMI **40**, as discussed in any of the schemes discussed above. If the coupler **14** can be identified (step **608**) in the image data **55**, the height $H_{\text{sub.c}}$ distance $D_{\text{sub.c}}$, and offset angle $\alpha_{\text{sub.c}}$ of coupler **14**, as identified in step **606**, can then be determined using the available image data **55** (step **606**) as discussed above, including using image processing routine **64**. As discussed above, image processing routine **64** can be programmed or otherwise configured to identify coupler **14** of trailer **16** within image data **55** (step **606**). In this manner, after the results of the initial scene scan (step **604**) are analyzed, controller **26** can determine if coupler **14** has been confirmed by the user (such as by way of HMI**40**). If coupler **14** has not been confirmed or if a determined coupler **14** has been rejected, the scene scan (step **604**) can be continued, including while instructing driver to move vehicle **12** to better align with trailer **16**, including by positioning the trailer **16** and/or coupler **14** within any of the above-described targets **45**, until coupler **14** is identified.

(62) When coupler **14** has been identified and confirmed, the path derivation routine **66** can be used to determine the vehicle path **32** to align hitch ball **34** with coupler **14** in step **610**. In this manner, the positioning $D_{\text{sub.h}}$, $\alpha_{\text{sub.h}}$ of coupler **14** is extracted from the image data **55** and used to place the coupler **14** within the stored data relating the image coordinates with the real-world coordinates of the area surrounding vehicle **12**. In doing so, controller **26** uses path derivation routine **66** to determine path **32** to align hitch ball **34** with the predicted position **28** of coupler **14** to an engaging position over hitch ball **34**, as described above with respect to FIGS. **20-23**.

(63) Once the path **32** has been derived, hitch assist system **10** can ask the user **U** to relinquish control of at least the steering wheel of vehicle **12** (and, optionally, the throttle **73** and brake, in the implementation of hitch assist system **10** described above wherein controller **26** assumes control of powertrain control system **72** and brake control system **70** during execution of operating routine **68**). When it has been confirmed that user **U** is not attempting to control steering system **20** (for example, using torque sensor **80**, as discussed above), controller **26** begins to move vehicle **12** along the determined path **32**. Hitch assist system **10** then controls steering system **20** (step **612**) to maintain vehicle **12** along path **32** as either user **U** or controller **26** controls the velocity of vehicle **12** using powertrain control system **72** and braking control system **70**. As discussed above, controller **26** or the user can control at least steering system **20**, while tracking the position $D_{\text{sub.c}}$, $\alpha_{\text{sub.c}}$ of coupler **14** until vehicle **12** reaches endpoint **35** (step **614**), wherein the vehicle **12** hitch ball **34** reaches the desired position **38a** for the desired alignment with coupler **14**, at which point operating routine **68** can end (step **618**), either by controlling brake system **70** to cause vehicle **12** to stop (which can be done progressively as vehicle **12** approaches such a point), or by issuing a command to the user to stop vehicle **12** (which can also be done progressively or by a countdown as vehicle **12** approaches the desired location) before deactivating hitch assist system **10**. Vehicle **12** can then be driven normally with system **10** remains idle until a reactivation input **620** is received, at which point the above-described method restarts at the scanning step **604**.

(64) It is to be understood that variations and modifications can be made on the aforementioned

structure without departing from the concepts of the present invention, and further it is to be understood that such concepts are intended to be covered by the following claims unless these claims by their language expressly state otherwise.

Claims

1. A vehicle hitching assistance system, comprising: a controller: acquiring image data from the vehicle; executing an initial alignment process, including: searching for a trailer, positioned past a minimum distance from the vehicle, within a preselected area within the image data, the preselected area being less than a total field of the image data in directions corresponding with both a longitudinal distance between the vehicle and the trailer, by an amount corresponding with a maximum detection distance, a minimum detection distance corresponding with a proportion of the trailer relative to the total field, and a lateral direction perpendicular to the longitudinal distance by an amount corresponding with a known steering limit of the vehicle; and when the trailer is not identified within the preselected area, presenting an indication to a driver of the vehicle to reposition the vehicle such that the trailer is within the preselected area and removing the indication when the trailer is identified within the preselected area, the trailer remaining past the minimum distance from the vehicle; and after completion of the initial alignment process, executing an automated backing process, including identifying a coupler of the trailer and outputting a steering signal to the vehicle to cause the vehicle to steer to align a hitch ball of the vehicle with the coupler of the trailer during reversing of the vehicle toward the trailer.
2. The system of claim 1, wherein the controller acquires the image data from an imaging system included with the vehicle, the imaging system having at least one camera, the total field of the image data corresponding with a total field of view of the at least one camera.
3. The system of claim 2, wherein: the controller outputs the steering signal to a steering system included with the vehicle; and the controller derives the steering signal based on at least a maximum steering angle of the steering system.
4. The system of claim 1, wherein the preselected area of the image data is a circular target area disposed at a preselected location in a central portion of the image data.
5. The system of claim 1, wherein the preselected area within the image data is defined by a designated boundary comprising respective portions based on the maximum detection distance, the minimum detection distance, and the known steering limit of the vehicle.
6. The system of claim 5, wherein the respective portions of the designated boundary are based on a correlation of the total field of the image data with an area of an assumed ground plane on which the vehicle is positioned visible within the total field.
7. The system of claim 6, wherein the area of the assumed ground plane includes: a maximum coupler detection distance corresponding with the resolution of the image data; a minimum trailer identification distance corresponding with the proportion of the trailer relative to the total field; and left and right maximum steerable paths extending from the vehicle in a reversing direction corresponding with the known steering limit of the vehicle.
8. The system of claim 1, wherein, during the initial alignment process, the controller further outputs a video image displayable on a human-machine interface within the vehicle including: the image data; a graphic overlay of the preselected area on the image data in a proportionally correlated manner; and a message indicating to the driver of the vehicle to reposition the vehicle until the trailer is identified within the preselected area.
9. The system of claim 8, wherein the controller begins executing the initial alignment process, including outputting the graphic overlay and message in the video image, upon activation of the system.
10. The system of claim 9, wherein, during the initial alignment process, the controller: receives an input from the human-machine interface corresponding with a user indication of a trailer within the

image data; and outputs the graphic overlay and message in the video image only after receiving the user indication of the trailer within the image data and failing to identify any trailer within the preselected area of the image data.

11. The system of claim 1, wherein the controller: during the initial alignment process, identifies the trailer within the preselected area of the image data and determines if a sensing condition and a visibility condition are met; executes the automated backing process, including identifying the coupler of the trailer and outputting the steering signal to the vehicle to cause the vehicle to steer to align the hitch ball of the vehicle with the coupler as a part of a first hitch assist mode when the sensing condition and visibility condition are not met; further implements a second hitch assistance mode when one of the sensing condition and the visibility condition are not met, the second hitch assistance mode including presenting an image of a best fit path to align the hitch ball with the coupler to a user for guiding user control of the vehicle in reversing the vehicle toward the trailer; receives a selection signal from the vehicle corresponding with a user selection of a mode before implementing either the first or second hitch assistance mode; and causes the vehicle to present an indication that the first hitch assist mode may not be selected when one of the sensing condition and the visibility condition are not met.

12. A method for assisting a vehicle in hitching with a trailer, comprising: acquiring image data for a field of view away from a rear of the vehicle; executing an initial alignment process, including: searching for a trailer, past a minimum distance from the vehicle, within a preselected area within the image data, the preselected area being less than a total field of the image data in directions corresponding with both a longitudinal distance between the vehicle and the trailer and a lateral direction perpendicular to the longitudinal distance; outputting a video image displayable on a human-machine interface within the vehicle including the image data, a graphic overlay of a target area, as a representation of the preselected area, on the image data in a proportionally correlated manner; and presenting an indication to a driver of the vehicle to reposition the vehicle when the trailer is not identified within the preselected area and removing the indication when the trailer is identified within the preselected area, the trailer remaining past the minimum distance from the vehicle; and after completion of the initial alignment process, executing an automated backing process, including identifying a coupler of the trailer and outputting a steering signal to cause the vehicle to steer to align a hitch ball of the vehicle with the coupler during reversing of the vehicle toward the trailer.

13. The method of claim 12, wherein the preselected area of the image data is within a designated boundary comprising respective portions based on a resolution of the image data, a proportion of the trailer relative to the total field, and a known steering limit of the vehicle.

14. The method of claim 13, wherein the respective portions of the designated boundary are based on a correlation of the total field of the image data with an area of an assumed ground plane on which the vehicle is positioned visible within the total field.

15. The method of claim 12, wherein the video image output during the initial alignment process further includes: a message indicating to the driver of the vehicle to reposition the vehicle until the trailer is identified within the preselected area.

16. A vehicle hitching assistance system, comprising: a controller: acquiring image data from the vehicle; searching for a trailer, positioned past a minimum distance from the vehicle, within a preselected area within the image data, the preselected area being less than a total field of the image data in directions corresponding with both a longitudinal distance between the vehicle and the trailer, by an amount corresponding with a maximum detection distance, a minimum detection distance corresponding with a proportion of the trailer relative to the total field, and a lateral direction perpendicular to the longitudinal distance by an amount corresponding with a known steering limit of the vehicle; determining that the trailer is not within the preselected area and executing an initial alignment process, including presenting an indication to a driver of the vehicle to reposition the vehicle such that the trailer is within the preselected area and removing the

indication when the trailer is identified within the preselected area, the trailer remaining past the minimum distance from the vehicle; and after completion of the initial alignment process, executing an automated backing process, including identifying a coupler of the trailer and outputting a steering signal to the vehicle to cause the vehicle to steer to align a hitch ball of the vehicle with the coupler of the trailer during reversing of the vehicle toward the trailer.

17. The system of claim 16, wherein the controller acquires the image data from an imaging system included with the vehicle, the imaging system having at least one camera, the total field of the image data corresponding with a total field of view of the at least one camera.

18. The system of claim 17, wherein: the controller outputs the steering signal to a steering system included with the vehicle; and the controller derives the steering signal based on at least a maximum steering angle of the steering system.

19. The system of claim 16, wherein the preselected area of the image data is a circular target area disposed at a preselected location in a central portion of the image data.

20. The system of claim 16, wherein, during the initial alignment process, the controller further outputs a video image displayable on a human-machine interface within the vehicle including: the image data; a graphic overlay of the preselected area on the image data in a proportionally correlated manner; and a message indicating to the driver of the vehicle to reposition the vehicle until the trailer is identified within the preselected area.
